

Data Collection and Preprocessing Phase

Date	2June2025
Name	Sagar Sunil Bhosale
Project Title	Restaurant Recommendation System
MaximumMarks	6 Marks

Data Preprocessing

The images will be preprocessed by resizing, normalizing, augmenting, denoising, adjusting contrast, detecting edges, converting color space, cropping, batch normalizing, and whitening data. These steps will enhance data quality, promote model generalization, and improve convergence during neural network training, ensuring robust and efficient performance across various computer vision tasks.

Section	Description
Data Overview	The dataset contains restaurant information from Zomato, including name, reviews, ratings, cuisines, cost, and more. The data is cleaned, deduplicated, and preprocessed for building a content-based recommendation system.
Resizing	<i>Not applicable for text data.</i>
Normalization	Ratings are normalized to a 1-5 scale using MinMaxScaler. Text is lowercased and punctuation is removed.
Data Augmentation	Not applicable for text data.
Denoising	Text is cleaned by removing new line characters and punctuation.
Edge Detection	Not applicable for text data.

ColorSpace Conversion	Not applicable for text data.
ImageCropping	Not applicable for text data.
BatchNormalization	Not applicable for text data.
DataPreprocessingCode Screenshots	
LoadingData	<pre> # Mounting Google Drive #from google.colab import drive #drive.mount('/content/drive') import csv # Specifying the path to the dataset file file_path = '/content/zomato.csv' # Reading the dataset into a Pandas DataFrame #df = pd.read_csv(file_path,encoding = 'ISO-8859-1', low_memory = False) df = pd.read_csv(file_path, encoding='ISO-8859-1', on_bad_lines='skip', engine='python') # Displaying the first few rows of the dataset to ensure it's loaded correctly df.head() </pre>
Resizing	<i>Not applicable</i>
Normalization	<pre> # Computing Mean Rating restaurants = list(df['name'].unique()) df['Mean Rating'] = 0 for i in range(len(restaurants)): df['Mean Rating'][df['name'] == restaurants[i]] = df['rate'][df['name'] == restaurants[i]].mean() #Scaling the mean rating values from sklearn.preprocessing import MinMaxScaler scaler = MinMaxScaler (feature_range = (1,5)) df[['Mean Rating']] = scaler.fit_transform(df[['Mean Rating']]).round(2) </pre>
Data Augmentation	<i>Not applicable</i>
Denoising	<pre> ## Lower Casing df["reviews_list"] = df["reviews_list"].str.lower() ## Removal of Punctuations import string PUNCT_TO_REMOVE = string.punctuation def remove_punctuation(text): """custom function to remove the punctuation""" return text.translate(str.maketrans('', '', PUNCT_TO_REMOVE)) df["reviews_list"] = df["reviews_list"].apply(lambda text: remove_punctuation (text)) </pre>
Edge Detection	<i>Not applicable</i>

ColorSpace Conversion	<i>Not applicable</i>
ImageCropping	<i>Not applicable</i>
BatchNormalization	<i>Not applicable</i>